

Stretching a single sugar chain

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ABSTRACT

Single molecule-stretching experiments have been extensively performed to evaluate the mechanical properties of biomacromolecules including nucleic acids and proteins but very few on polysaccharides. Polygalacturonic acid (PGA), a major pectin component in plant cell walls, plays key roles in plant physiology. In this study, an assay was developed to stretch PGA molecules by single-molecule experiments. The aldehyde of the reduced end of PGA was attached to an aminosilane-coated glass while the other end was biotin-functionalized and attached to a streptavidin-coated microbead. The elasticity of PGA was investigated under wide solution conditions. The persistence length (P) of PGA decreased from 5.8 to 2.8 nm as the concentration of monovalent cations was increased to 4 M. Notably, increasing the concentration of divalent cations first decreased P from 5.8 to 0.97 nm then increased P to 2.1 nm, with a reversal around 200 mM, matching zeta-potential evidence of charge neutralization then charge reversal. Simulations results revealed that Ca^{2+} formed bidentate coordination and induced significant cross-linking of PGA, whereas Mg^{2+} , due to its strong hydration shell, bound in a monodentate manner and induced softening but not cross-linking of PGA.

1. Introduction

Nucleic acids, proteins, and polysaccharides represent the three fundamental classes of biological macromolecules. Substantial progress has been made in understanding the structure, dynamics, and mechanical properties of nucleic acids and proteins at the single-molecule level, revealing critical insights into DNA/RNA folding, transcription, and protein conformational changes (Fang et al., 2023; Haghizadeh et al., 2023; Kriegel et al., 2017; Tapia-Rojo, 2025; Tapia-Rojo et al., 2023; Wilson & Corrêa, 2025). Extensive studies have demonstrated that the structural and electrostatic properties of nucleic acids and proteins are modulated by divalent cations such as Ca^{2+} and Mg^{2+} . While Mg^{2+} typically stabilizes macromolecular conformations through outer-sphere interactions, Ca^{2+} more readily forms inner-sphere complexes, inducing cross-linking, aggregation, or precipitation (Collins, 2006). These effects can lead to local charge neutralization or even reversal, promoting phase transitions such as condensation or crystallization (Huynh et al., 2016). Our previous work revealed that increasing the concentrations of divalent cations causes a charge-inversion of DNA and thus a decrease in the melting temperature (Zhang et al., 2025). Charge inversion induced by

divalent cations has also been observed in bacterial membranes and biological ion channels (García-Giménez et al., 2010; Kim & Sung, 2005; Luan et al., 2016). These ion-specific interactions underscore the importance of ionic conditions in modulating the stability, structure, and assembly of nucleic acids and proteins in both physiological and experimental contexts. In contrast, although polysaccharides are biological macromolecules with structural and functional significance, they have received comparatively little attention in single-molecule studies, leaving their mechanical behavior and molecular interactions largely unexplored. Polysaccharides are present in nature in large abundance (Peesapati & Roy, 2023). They have incredible film-forming abilities, also being the primary structural scaffold in plant and animal exoskeletons (Díaz-Montes, 2022). They are often exposed to different mechanical stresses, which can cause significant deformations of the sugar residues. For instance, during the lysozyme reaction in bacterial cell walls, the enzyme forces the sugar residues into a highly strained conformation (Lee et al., 2004). The response of polysaccharides to external forces has been examined by stretching single polysaccharides using an atomic force microscope (AFM) (Lee et al., 2004; Haverkamp et al., 2005; Marszałek & Dufrêne, 2012; Marszałek et al., 2003;

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Peesapati & Roy, 2023; Rief et al., 1997; Zhang & Marszalek, 2006). These measurements have expanded the conformational analysis of ring structures to include force-induced chair-boat and chair inversion transitions of the sugar residues. For example, Marszalek et al. (2003) observed force-induced conformational transitions in heparin and postulated that these conformational changes may finely regulate the affinity of various ligands toward heparin in secretory granules or in the extracellular matrix.

Polymeric molecules based on sugar residues give rise to a diverse array of polysaccharides, with their names determined by the types of monomers and glycosidic bonds. Notable examples include cellulose, pectin, chitosan, hyaluronic acid, and chondroitin sulfate (Tanhatan-Nasseri et al., 2011). Among various polysaccharides, pectin is a heterogeneous compound that accounts for approximately 35 % of the dry weight of most plant cell walls and plays important roles in plant physiology, including regulating cell growth and triggering defense mechanisms (Wang et al., 2018). Polygalacturonic acid (PGA), a major pectin component in plant cell walls, has a linear backbone composed of galacturonic acid residues connected by α -(1, 4)-glycosidic linkages (Mohandas et al., 2023; Ralet et al., 2008; Tanhatan-Nasseri et al., 2011). PGA has been applied in biodegradable films, drug delivery, food gelling agents and stabilizers, and electrospun fibers (Cataldo et al., 2012; Gupta et al., 2019; Mendes et al., 2017). These functions are closely related to its mechanical properties, including elasticity and force-induced conformational transitions (Lee et al., 2004; Marszalek et al., 2003; Marszalek & Dufrêne, 2012). The biomechanical properties of individual PGA molecules have previously been studied using AFM in the high-force regime, by simply relying on physical adsorption to facilitate fast stretching of the PGA between surfaces (Williams et al., 2007). Single-molecule studies of PGA are challenging due to the lack of reliable end-functionalization techniques. Being able to functionalize both ends of the PGA will enable the investigation of the mechanical properties using techniques such as optical tweezers or magnetic tweezers (MT).

Single-Molecule Force Spectroscopy (SMFS) has emerged as a pivotal method in molecular biophysics and mechanochemistry, enabling precise probing of the mechanical properties and molecular interactions of polymers and biomolecules at the single-molecule level (Cheng & Cui, 2015; Liu & Zhang, 2012; Marszalek & Dufrêne, 2012; Ott et al., 2017; Wilson & Corrêa, 2025; Yang et al., 2020). MT, as a robust and versatile SMFS platform, offer unique capabilities for exploring the mechanics of biomolecules under controlled force conditions. MT enables the application of constant forces over extended durations, making it particularly well-suited for investigating slow, reversible conformational changes and kinetic processes in biomolecular systems (Fang et al., 2023; Tapia-Rojo, 2025). Moreover, MT allows for the parallel manipulation of multiple single molecules, facilitating high-throughput data collection and statistically meaningful analyses. These strengths have increasingly been harnessed to study the mechanical properties of nucleic acids, proteins, and polysaccharides. Mohandas et al. (2023) proposed a method to create a DNA-PGA-DNA composite chain by inserting a PGA molecule between two biotinylated DNA double-strand handles using bivalent streptavidin linkers, which could potentially be applied in single-molecule optical tweezers experiments. Since the composite chain contains DNA double-strand handles, the resulting force-extension curves correspond to the integrated response of the entire construct, rather than exclusively to the PGA chain.

The interaction between cations and PGA has broad applications in the food, pharmaceutical, and pollutant treatment industries (Camci-Unal & Pohl, 2010; Itoh et al., 2007; Liu et al., 2003). Due to its multiple carboxylate-binding sites, PGA can act as a metal ion sequestering agent (Cataldo et al., 2012). The conformational features of PGA are very likely involved in its chemical and biochemical properties. These include interactions with positively charged ions, in particular, divalent cations (for example Ca^{2+}), which promote secondary structure formation (Cescutti & Rizzo, 2001). The effects of PGA's chemical structure,

concentration, and binding strength of Ca^{2+} have been studied using various physicochemical methods such as potentiometry, spectrophotometry, viscometry, and calorimetry (Malovíková et al., 1994; Morris et al., 1982). Despite the great interest in these processes, there is still little research on the effects of cations on the mechanical properties of PGA and the changes in the conformation of PGA induced by cations at the single-molecule level, which is essential for elucidating the structural stability, gelation behavior, and intermolecular interactions of PGA.

In this paper, we have developed a novel MT assay enabling direct and specific tethering of PGA, allowing precise and long-term measurements under a wide concentration range. Previous AFM-based studies only examined PGA using nonspecific adsorption, which lacked directional control and long-term stability. This work improves upon prior efforts by introducing a two-end functionalization strategy, ensuring high-resolution and reproducible stretching. Our two-end functionalization strategy is broadly applicable to other linear carbohydrate polymers. Moreover, this platform allows the investigation of phenomena such as single-chain charge reversal, which are difficult to access using conventional ensemble-based methods, due to chain condensation in bulk systems. Combined with zeta potential measurements and molecular dynamics (MD) simulations, our assay enables mechanistic dissection of how mono- and divalent cations alter PGA elasticity, charge state, and conformational dynamics. Notably, we report two previously inaccessible phenomena at the single molecule level: (i) ion-induced charge inversion of PGA upon divalent cation binding, and (ii) Ca^{2+} -specific crosslinking via bidentate coordination with carboxyl groups of PGA, in contrast to the monodentate, non-crosslinking behavior of Mg^{2+} . These findings provide a molecular-level framework for understanding ionic control of polysaccharide conformation and mechanics, with implications for plant cell wall physiology, food hydrocolloid behavior, and the design of ion-responsive biomaterials.

2. Materials and methods

2.1. Materials

The sodium salt of PGA (MW 250, 000-1,000, 000, Shanghai Yuanze Bio-Technology Co., Ltd., Shanghai, China), pectate lyase *Aspergillus* sp. (Megazyme-Food Tech Solutions Ltd., New Zealand), 3-Aminopropyltriethoxysilane (APTES, Shanghai Macklin Biochemical Technology Co., Ltd., Shanghai, China), 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropionophenone (Irgacure 2959, Beijing Solarbio Science & Technology Co., Ltd., Beijing, China), sulfo-NHS-biotin (Shanghai Aladdin Biochemical Technology Co., Ltd., Shanghai, China), cysteamine (Beijing Solarbio Science & Technology Co., Ltd., Beijing, China), Decon 90 (Decon Laboratories Ltd., Hove, UK) were purchased and used without further purification. The stock solutions of NaCl, KCl, MgCl_2 and CaCl_2 were prepared using a buffer solution containing 5 mM Tris-HCl at pH 8.0.

2.2. Activation of non-reducing end of PGA

We dissolved 1 mg of PGA in 1 mL of 100 mM Tris-HCl, then mixed with 4 μL pectate lyase (diluted 1:1000 from stock solution, the working concentration was 1.4 U/mL), and incubated at 50 °C for three days. The optimization of reaction temperature and reaction time was confirmed by the results presented in Fig. S1. Enzyme activity was terminated by boiling for 3 min. The mixture was then precipitated with 70 % ethanol, followed by centrifugation (12,000 rpm for 2 min) (5424R, Eppendorf SE, Hamburg, Germany). After removing the supernatant, the precipitate was re-dissolved in 100 mM phosphate buffer solution (PBS, pH 7.5).

2.3. Linking the reducing end of PGA to the bottom coverslip of the flow channel

The upper and bottom coverslips (24×40 mm, 24×60 mm, Citotest Scientific Co., Ltd., Nanjing, China) were treated with 2 % Decon 90 and piranha solution, then ultrasonically cleaned. The upper coverslip was dried for later use. The bottom coverslip was then immersed in APTES solution (one mL of APTES in 50 mL of absolute ethanol) and incubated for 2 h. During this process, APTES formed covalent bonds with the hydroxyl groups on the surface of the bottom coverslip (Fig. S2) (Huang et al., 2012). As shown in Fig. S3A, the upper and bottom coverslips were assembled using double-sided adhesive tapes to form flow channels. Before adding PGA into the channels, amino-functionalized microspheres (2.2 μ m, Spherotech Inc., Chicago, USA) diluted in 100 mM PBS were introduced to serve as reference beads to eliminate drift of the stage during the experiment. Briefly, each channel was rinsed with 200 μ L diluted microspheres. The beads were incubated for 20 min. Then, the channels were rinsed with 500 μ L of 100 mM PBS to remove any free beads. Secondly, 200 μ L of the solution of PGA with activated non-reducing end was introduced into each channel and incubated for 30 min, followed by rinsing with 500 μ L of 100 mM PBS. The aldehyde group at the reducing end of PGA reacted with the aminoxy group on the surface of the APTES-coated bottom coverslip, forming a stable oxime linkage (Mohandas et al., 2023) (Fig. 1).

2.4. Biotin functionalisation at the non-reducing end of PGA

According to the method of Mohandas et al. (2023), cysteamine and sulfo-NHS biotin ester were mixed in a 1:5 M ratio in 100 mM PBS (pH 7.5) at room temperature for 3 h to generate thiol-NHS-biotin linker (Fig. S3B). The thiol-NHS-biotin linker and 0.1 % (w/v) Irgacure 2959 were dissolved in 200 μ L PBS and introduced into the PGA-anchored channels. Next, the channels were irradiated with UV light (254–366 nm) for 30 min. After potential linkage formation, the channels were rinsed with 500 μ L PBS to remove free thiol-NHS-biotin and the photo-

initiator. Finally, the streptavidin-coated superparamagnetic microbeads (M-270, Dynabeads, Thermo Fisher Scientific Inc., Waltham, USA) were diluted in PBS and introduced into the channels, allowing them to bind to the biotin-labeled end of the PGA. After incubation for 20 min, unbound beads were removed by gentle rinsing with 500 μ L PBS.

2.5. The elasticity measurements of PGA using MT

In the PGA elasticity measurements, one end of a PGA molecule was anchored to a superparamagnetic microbead, and the other end was anchored to the bottom coverslip (Fig. 1). A pair of magnets was used to apply forces to the anchored PGA molecule. At low forces, a long holding time was required to obtain an accurate average extension of PGA. We found that 30 s was sufficient within our force range (Fig. S4). Thus, we hold PGA for 30 s at each constant force to calculate the average extension. To determine the exact applied force (F) at each magnet height (h), we fitted F as a function of h for each paramagnetic bead (Fig. S5). The platform drift during measurements was eliminated by subtracting the displacement of the reference bead from that of the magnetic bead (Fig. S6). Fig. S7 shows a typical force-extension curve (FEC) obtained from PGA in 150 mM NaCl. More details about MT can be found in our previous studies (Chen & Zhang, 2024).

2.6. Determination of zeta potential and particle size distribution of PGA

The zeta potential and particle size distribution of PGA were determined using a laser diffraction particle size analyzer (Nano ZS90, Malvern Instruments Ltd., UK) at 25 $^{\circ}$ C. The solutions of PGA (0.01 mg/mL) were diluted with a 5 mM Tris-HCl buffer with different concentrations of cations.

2.7. Molecular dynamics simulation

The simulations were performed using GROMACS 5.1.2 software, with the CHARMM36 force field (Berendsen et al., 1995; Huang et al.,

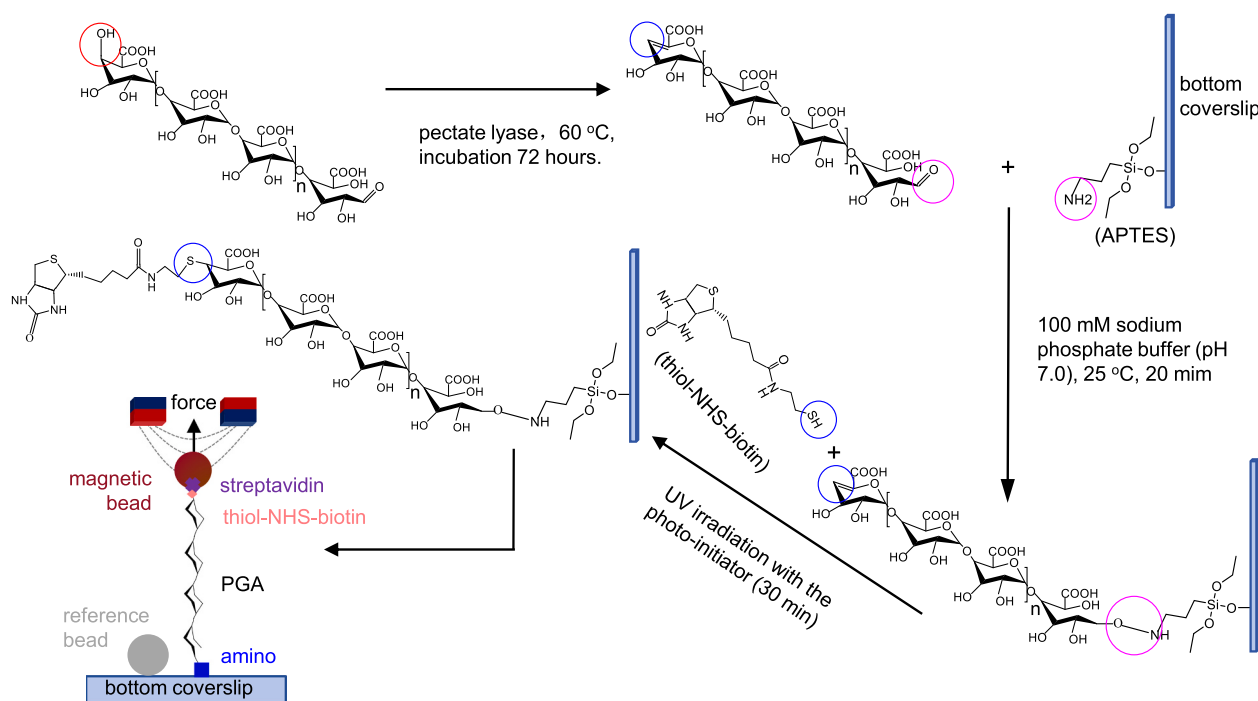


Fig. 1. The flow chart of end-functionalized modification PGA sample for single molecule stretching. The red circle represented the activation site of the non-reducing end of PGA. The magenta circle represented the reaction site where the reducing end of PGA was connected to the bottom coverslip of the flow channel. The blue circle represented the reaction site of biotin-functionalized of non-reducing end of PGA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2017). The sequence of the PGA (Gal_{10}), connected together via α -(1, 4)-glycosidic linkages, was chosen in simulations, which was generated by CHARMM-GUI using Glycan Reader & Modeler mode (Jo et al., 2008; Lee et al., 2016). Initially, PGA was placed in a simulation box (Martínez et al., 2009). To determine the average hydration number and the structure of the hydration shell of cations, systems containing PGA and four cations (Na^+ , K^+ , Mg^{2+} and Ca^{2+}) were simulated, with only one cation present in each system. For comparison, we also performed independent simulations of the cations alone in pure water, each system likewise containing only one cation. In order to clarify Ca^{2+} induces cross-linking of PGA, the simulations of two and four PGA chains in the presence of different concentrations of Ca^{2+} were performed. The composition of the systems in the simulations is shown in Tables S1–4. Subsequently, each simulation system was solvated with TIP3P water molecules and neutralized with a certain number of ions corresponding to the actual salt concentration in MT experiments. Energy minimization was first performed to prevent atomic clashes using steepest descent algorithm, followed by equilibration at 300 K in NVT ensemble and at 1 bar in NPT ensemble. LINCS was applied to constrain all bonds connecting the hydrogen atoms (Hess et al., 2008). Following pre-equilibration, all MD simulations were carried out for 50 ns. PyMOL v2.3.0 (Schrödinger, California, USA) was used for visualization, and g_mmpbsa was used to calculate the binding energies between cations and PGA (Kumari et al., 2014).

3. Results and discussion

3.1. The elasticity of PGA under various salt conditions

By functionally modifying the terminal ends of PGA, we successfully applied it into the MT system. The length distribution of PGA was predominantly between 450 and 3200 nm (Fig. S8). Fig. 2A and B show the FECs of PGA measured at different concentrations of monovalent cations (Na^+ , K^+). The extension of PGA decreased with increasing cations

concentrations at the same force. This phenomenon was also observed in the presence of divalent cations (Mg^{2+} , Ca^{2+}) at low concentrations, as shown in Fig. 2C and D. Interestingly, when the concentrations of Mg^{2+} or Ca^{2+} exceeded 200 mM, further increases in divalent cations concentrations resulted in an unexpected increase in the extension of PGA. To describe this phenomenon more clearly, we compared the changes in PGA extension with cations concentrations under constant forces (10 pN, 6 pN, 4 pN, 2 pN, and 1 pN) (Fig. 2E–H). The extension of PGA consistently decreased with increasing monovalent cations concentrations, even at 1 pN with 4 M monovalent cations concentration (Fig. 2E and F). The decrease in extension of PGA was likely due to the charge neutralization by cations. In the presence of Mg^{2+} , a noticeable extension reversion of PGA was observed when the concentrations of Mg^{2+} exceeded 200 mM. For Ca^{2+} , this extension reversion appeared at the same concentration. The reason for this phenomenon might be that the surface charges of PGA were fully neutralized by Mg^{2+} or Ca^{2+} at 200 mM, thus minimizing electrostatic repulsion between different PGA segments and resulting in the shortest extension. With the further increase in divalent cations concentrations, additional Mg^{2+} or Ca^{2+} binding to PGA led to a reversal of the surface charges of PGA, which enhanced electrostatic repulsion between PGA segments and ultimately increased the extension of PGA. Notably, the extension of PGA did not sharply increase beyond the inversion concentration of Ca^{2+} (200 mM) compared to Mg^{2+} under the constant force of 1 pN (Fig. 2G and H). In addition, we measured the variation in PGA extension under constant forces at divalent salt concentrations of 100 mM and 200 mM (Figs. 3 and S9). We observed aggregation of a single PGA chain at 2.2 pN when the concentration of Ca^{2+} was 200 mM, whereas no aggregation was observed at forces larger than 1 pN under other salt conditions. These results indicated that, unlike Mg^{2+} , Ca^{2+} induced the aggregation of PGA at 200 mM, requiring greater force (approximately 40 pN) to destroy. This behavior is likely attributed to the ability of Ca^{2+} to crosslink PGA chains under low-force conditions, whereas Mg^{2+} appears to only soften the PGA chains without crosslinking.

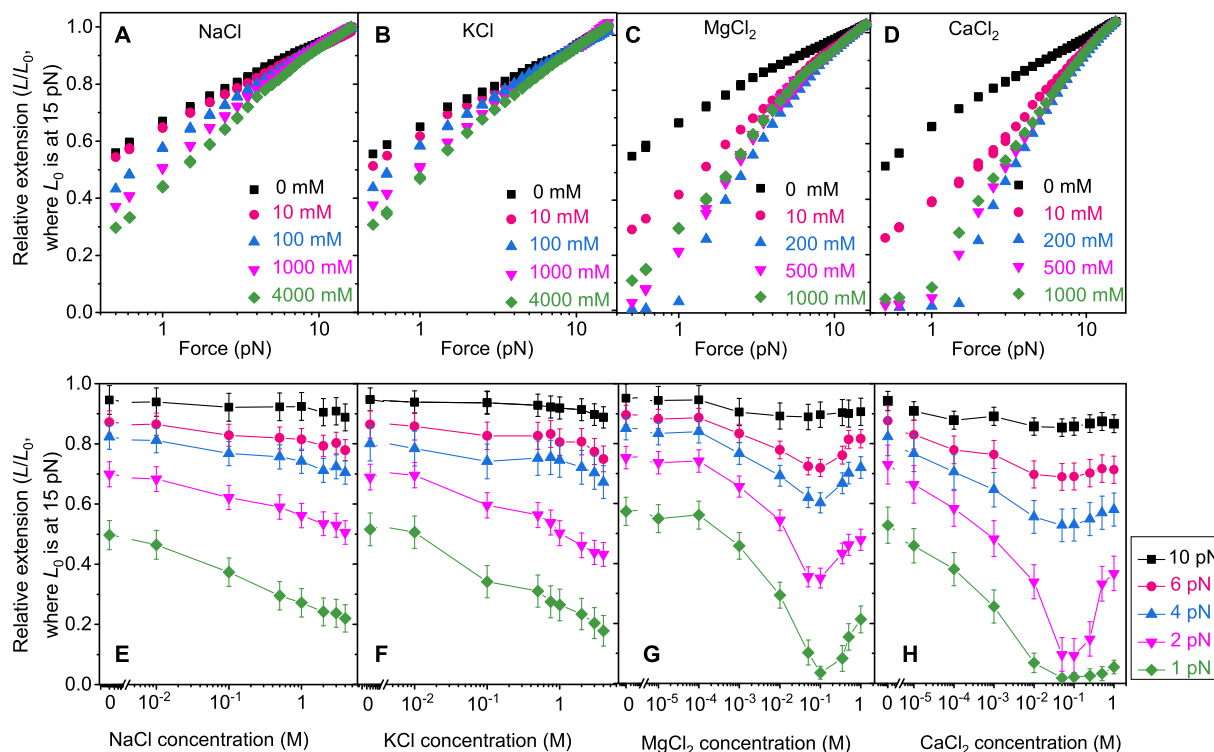


Fig. 2. The force-extension curves of PGA at different concentrations of cations. The relative extension was calculated, with the extension of PGA at 15 pN chosen as the reference extension (L_0). (A) NaCl. (B) KCl. (C) MgCl_2 . (D) CaCl_2 . The changes in the extension of PGA as a function of cations concentrations at different constant forces. (E) NaCl. (F) KCl. (G) MgCl_2 . (H) CaCl_2 .

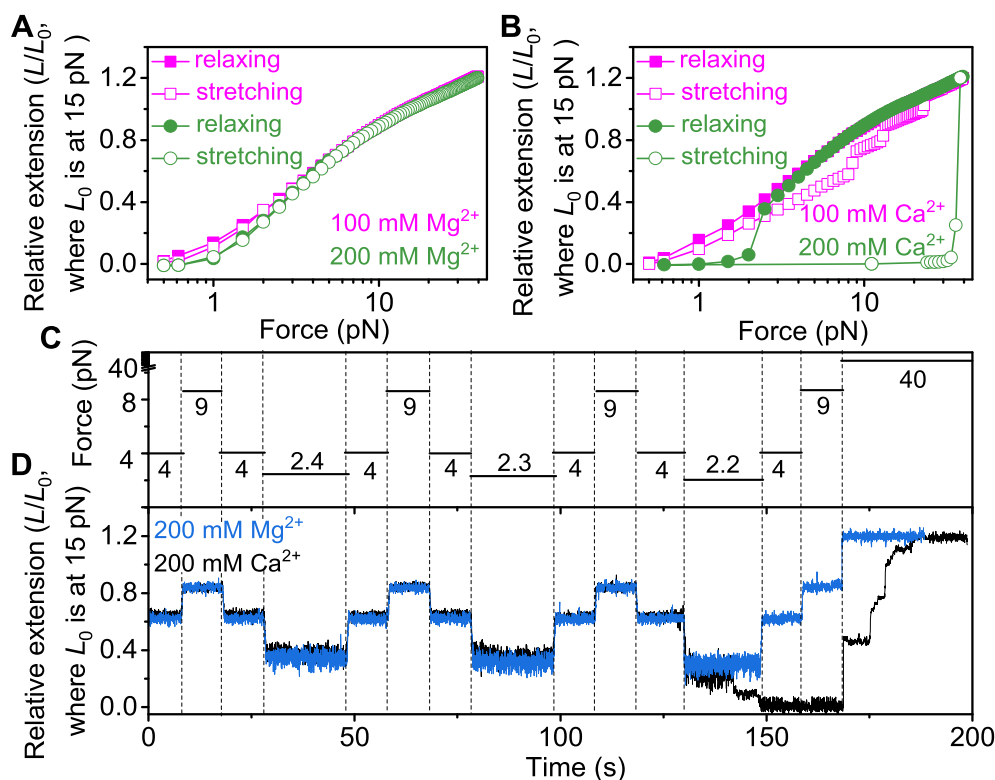


Fig. 3. Salt-induced aggregation of PGA. (A) Force-extension curves of PGA in the presence of 100 mM and 200 mM Mg^{2+} during both stretching and relaxing processes. (B) Force-extension curves of PGA in the presence of 100 mM and 200 mM Ca^{2+} during both stretching and relaxing processes. (C) The repeating force-cycles were used to determine the condensation forces. (D) The typical time-extension course recorded during repeating force-cycles at 200 mM MgCl_2 and 200 mM CaCl_2 .

To describe the changes in the elastic properties of PGA in the presence of cations, we fitted the FECs to the worm-like-chain (WLC) model using the Marko-Siggia interpolation formula (Marko & Siggia, 1995). The FECs of PGA were generated by subjecting them to a series of gradually increasing constant forces during the stretching process and a series of decreasing constant forces during the relaxing process. To avoid

the effects of Ca^{2+} -induced crosslinking of PGA on the fitting results, we used only the data obtained during the relaxing process for WLC fitting:

$$F = k_B T / P \left[0.25 (1 - L/L_c)^{-2} + L/L_c - 0.25 \right] \quad (1)$$

The measured variables, L and F , represent the molecular extension

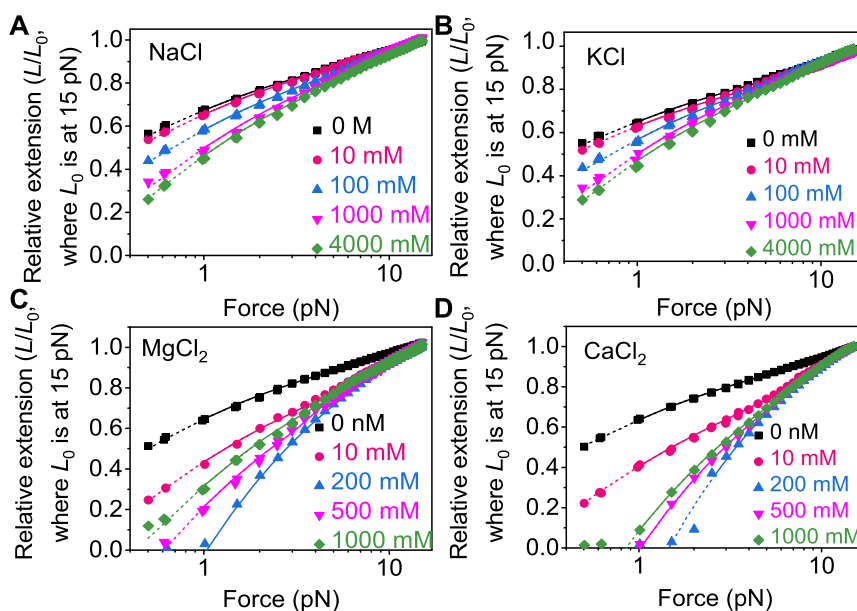


Fig. 4. Typical force-extension curves and WLC fitting curves of PGA at different salt conditions. The relative extension was calculated, with the extension of PGA at 15 pN chosen as the reference extension (L_0). Experimental data were represented by different symbols and fitted curves depicted by solid lines. The dashed lines represent the extrapolated portion of the fitted curves. (A) NaCl. (B) KCl. (C) MgCl_2 . (D) CaCl_2 .

and the applied tensile force, respectively. The two fitted elastic parameters are the contour length (L_c) and the persistence length (P).

Figs. 4 and S10–S13 show the FECs of PGA at different concentrations of monovalent and divalent cations. At 100–200 mM Mg^{2+} and 100–500 mM Ca^{2+} , the extension of PGA decreased to about zero at forces even lower than 1 pN (Figs. 4 and S12–13), making it unsuitable for WLC fitting at these low forces. Notably, the fitting curves are in agreement with the extensions obtained from experiments at forces below 1 pN under other monovalent salts and low divalent salts concentrations (dashed lines). Nevertheless, to maintain methodological consistency and eliminate low-force artifacts, we selected the 1–15 pN force range for WLC fitting across all salt conditions. An exception is at 200 mM Ca^{2+} , to avoid PGA condensation, we fitted the data in the force range of 2.3 to 15 pN. In Figs. 5A–B and S14, we reported the values of P and L_c (Tables S5 and S6). The L_c of PGA remained almost constant across the investigated salt concentration ranges, regardless of the ion type or the molecular length of PGA (Fig. S14). Importantly, we determined that the P value of PGA under physiological salt conditions (150 mM NaCl) was 3.62 ± 0.54 nm (Fig. 5A). As shown in Fig. 5A, the P gradually decreased from 5.8 nm to 2.8 nm as the concentration of monovalent cations (Na^+ , K^+) increased from 0 mM to 4 M. Similarly, in the presence of divalent cations (Mg^{2+} , Ca^{2+}), a comparable decrease in the P values from 5.8 nm to 0.97 nm was observed as the divalent cations concentrations increased to 200 mM (Fig. 5B). Beyond this concentration, the P values increased from 0.97 nm to 2.1 nm as the divalent cations concentrations increased up to 1 M. These results indicated that divalent cations first soften and then re-stiffen PGA through charge neutralization then charge inversion. In contrast, monovalent cations soften PGA by charge neutralization.

3.2. The changes of zeta potential and particle size of the PGA at various cations concentrations

To verify cation-induced charge reversal of PGA, we measured the zeta potential and particle size of PGA-cation complexes under varying concentrations of mono- and divalent cations. As shown in Fig. 5C, the

initial zeta potential of PGA-cation complexes was -33.2 mV. With the addition of cations, the zeta potential of PGA-cation complexes became less negative. Specifically, the zeta potential of PGA changed monotonically from -33.2 mV to -1.65 mV in the presence of Na^+ at concentrations ranging from 0 M to 4 M, and from -33.2 mV to -3.81 mV in the presence of K^+ over the same concentrations range. Unlike monovalent cations, the zeta potential of the PGA-Mg complex approached zero when the Mg^{2+} concentration reached 200 mM. With a further increase in Mg^{2+} concentrations, the PGA-Mg complex acquired a positive net charge, reaching 7.73 mV at 1 M Mg^{2+} . The neutral zeta potential of PGA-Mg complex indicated that Mg^{2+} neutralized the negative charge of PGA. Beyond the neutral charge point, further increases in Mg^{2+} concentration reversed the zeta potential of the PGA-Mg complex, increasing electrostatic repulsion and molecular extension of PGA, as observed in our FECs analysis (Fig. 2C and G). Notably, different from the case of Mg^{2+} , in the presence of Ca^{2+} , a plateau phase (0.01 M–0.2 M) with zeta potential values close to zero was observed for the PGA-Ca complex. A similar phenomenon was reported by Larson et al. (2023), who studied the effects of aluminum ions on the zeta potential of exopolysaccharides from *Rhizobium tropici*. The appearance of the plateau phase of zeta potential might be attributed to Ca^{2+} -mediated cross-linking between PGA molecules, which led to rapid molecular aggregation and the formation of an egg-box structure (Fig. 3D) (Braccini & Pérez, 2001; Huynh et al., 2016; Larson et al., 2023). Our results agree with the following picture: At 10 mM Ca^{2+} , PGA chains aggregate into particulate structures, as indicated by the increase in particle size (Fig. 5D). At this stage, the surface zeta potential reaches zero (Fig. 5C). As the concentration of Ca^{2+} increases further, the electrostatic interactions between neutral PGA particles and free Ca^{2+} in solution remain weak, resulting in minimal changes to surface charge. Thus, over a range of concentrations, the zeta potential remains near zero. However, beyond 200 mM Ca^{2+} , MT experiments reveal an inversion of the persistence length of a single PGA chain (Fig. 5B), which coincides with a reversal in the surface charge of PGA particles (Fig. 5C) and a reduction in particle size (Fig. 5D). To better mimic the experimental conditions where PGA forms cross-linked particles composed of multiple

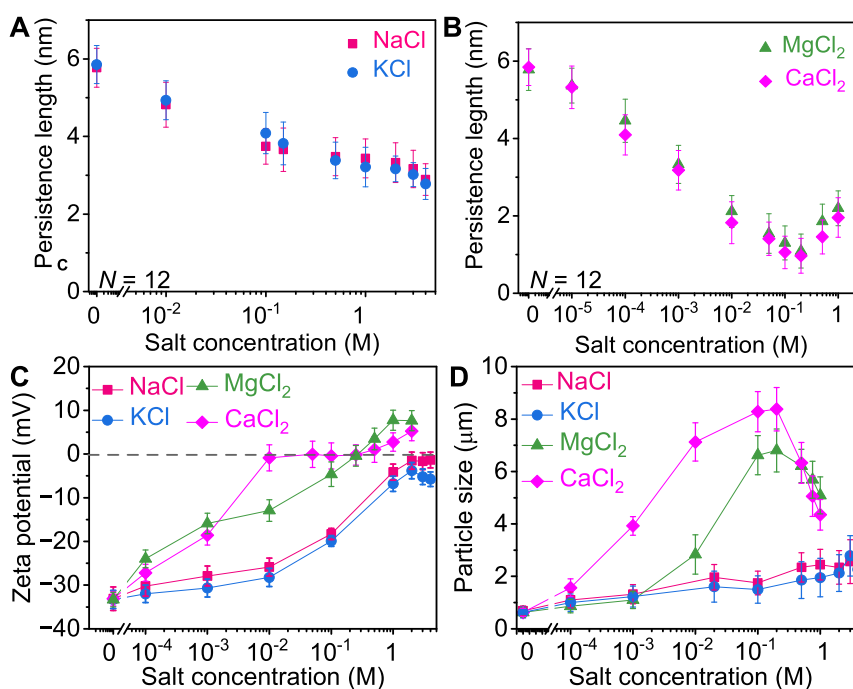


Fig. 5. The persistence length, zeta potential and particle size of PGA as functions of salt concentrations. (A–B) Fitted parameters as functions of salt concentrations. The plot displayed data points with error bars represented the average and standard deviation of the fitting parameters obtained from twelve molecules. (C–D) The zeta potential and particle size of PGA as functions of salt concentrations obtained from three experiments.

chains, we conducted molecular dynamics simulations using systems with two and four PGA chains at various Ca^{2+} concentrations. When the data are averaged per chain in these cross-linked systems, we observe that the number of bound Ca^{2+} ions tends to flatten out compared to that of a single chain as the number of PGA chains increases (Fig. S15). The average number of Ca^{2+} bound on per PGA chain decreased with increasing in the number of cross-linked chains in MD simulations (Fig. S15). This may be attributed to the partial burial of carboxyl groups within the interior of the formed particles, where a lower number of Ca^{2+} or a lower Ca^{2+} concentration was sufficient to neutralize the surface charge of PGA particles. In contrast, Mg^{2+} only induced intra-molecular compaction without cross-linking of PGA, resulting in a gradual reduction in zeta potential. Donati et al. (2009) also reported that Mg^{2+} -alginate solutions do not undergo gelation due to the weak interactions between Mg^{2+} and alginate. The mechanism is further supported by the changes in particle size of PGA-cation complexes with the increase in cations concentrations (Fig. 5D). In particular, the particle size of PGA-cation complexes increased with divalent cations, peaked at 200 mM, and then decreased as the cations concentrations increased. The particle size of the PGA-Ca complex was greater than that of PGA-Mg complex.

Overall, among monovalent cations at the same concentration, Na^+ was more effective than K^+ in reducing the zeta potential of PGA, which might be attributed to the difference in ionic radius. In the presence of 200 mM divalent cations, the zeta potential of PGA was nearly zero and the particle size of PGA was at its maximum. Further increases in concentrations of divalent cations resulted in a positive zeta potential and a decrease in particle size, indicating that divalent cations induced a reversal of the net charge of PGA (Fig. 5C and D). No reversal was observed in the net charge of PGA, even at monovalent cations concentrations as high as 4 M. A plateau phase in the zeta potential of PGA-Ca complex was observed, indicating that Ca^{2+} induced cross-linking between multiple PGA molecules (Fig. 5C). In contrast, Mg^{2+} primarily compacted single PGA molecules rather than cross-linking. However, in the single-molecule magnetic tweezers system, where the PGA molecules were constantly stretched, no cross-linking occurred between

PGA molecules when the force exceeded 1 pN, except at 200 mM Ca^{2+} (Fig. 2G and H). Therefore, the significant decrease in P values and extension of PGA was primarily due to charge neutralization induced by cations.

3.3. Binding characteristics of cations to PGA

To explore the interaction mechanisms between different cations and PGA at the atomistic scale, we performed all-atom MD simulations of PGA in different salt concentrations. The root mean square deviation (RMSD) and root mean square fluctuation (RMSF) of PGA were converged during our 50 ns production run and showed small fluctuations with respect to the average value, suggesting our MD simulations were converged and PGA maintained its canonical structure in each simulation system (Figs. S16–S23). We calculated the number of cations in the first shell of PGA using the threshold (Fig. 6A), r , which represents the distance between the center of mass of PGA and the first shell obtained from the first peak of the radial distribution function (RDF) of cations around PGA (Fig. S24). Our results show no significant difference in the number of cations between Na^+ and K^+ , as well as between Mg^{2+} and Ca^{2+} . The binding free energy between cations and PGA also showed similar results (Fig. 6B), which might be attributed to the fact that the effect of cations on PGA depends primarily on their valence state when cations-induced cross-linking of PGA were not involved. These results were consistent with the analysis of the P of PGA (Fig. 5B). A similar result was reported by Guilbaud et al. (2019), who found that the metallic ion type does not influence the P of DNA.

We then focused on the interactions between cations and the carboxyl groups of PGA, which could be the key factor for inter-chain associations (Braccini & Pérez, 2001; Huynh et al., 2016; Larson et al., 2023; Lerbret & Assifaoui, 2022). We calculated the duration time of cations at the carboxyl groups of PGA. As shown in Fig. 6C, Ca^{2+} exhibited a significantly longer duration time compared to the other cations, which might be the reason that Ca^{2+} could mediate the cross-linking between PGA molecules, leading to the formation of an egg-box structure (Braccini & Pérez, 2001; Huynh et al., 2016; Larson

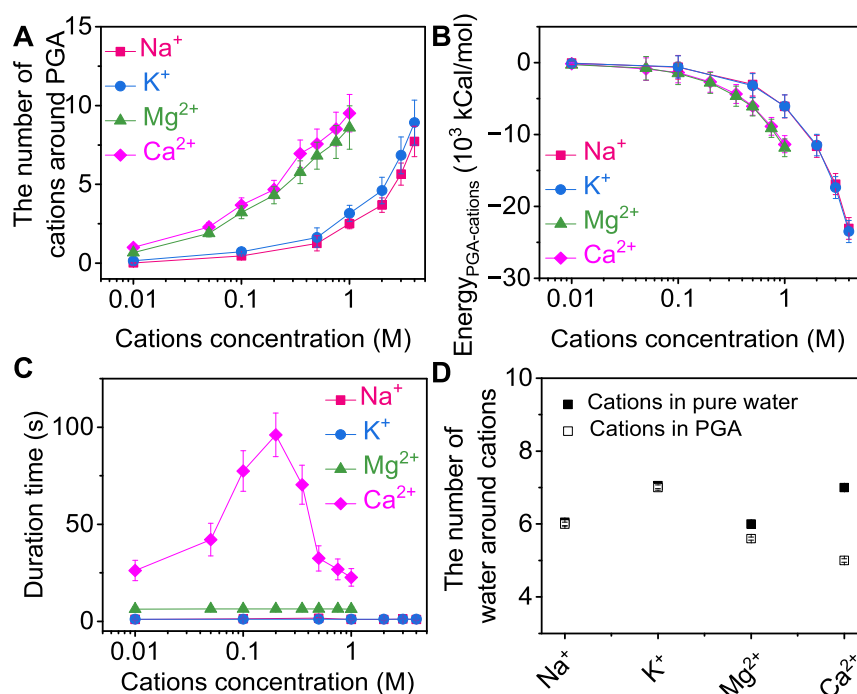


Fig. 6. (A) The number of cations in the first shell of PGA at different cations concentrations. (B) The binding free energy between PGA and cations at different cations concentrations. (C) The duration time of the cations in the first coordination shells of the carboxyl groups of PGA at different cations concentrations. (D) The number of water molecules in the first coordination shell of cations in pure water and in PGA systems, with each system containing only one cation.

et al., 2023). The duration time of Ca^{2+} reached a maximum at a concentration of 200 mM and decreased subsequently (Fig. 6C). These results indicated that the surface negative charges of PGA could be efficiently neutralized by strongly bound of Ca^{2+} at a concentration of 200 mM. The electrostatic screening effect of Ca^{2+} was manifested by the change in P observed from our MT experiments where the values of P reached the minimum around 200 mM Ca^{2+} (Fig. 5B). With further increases in Ca^{2+} concentration, continuous binding of Ca^{2+} to PGA repels Ca^{2+} on the carboxyl sites, which facilitates the process of association and dissociation of Ca^{2+} on the carboxyl sites and leads to the reduction in duration time. In addition, Mg^{2+} exhibited shorter duration time, which may be due to the fact that Mg^{2+} interacted with PGA as “diffusely bound” counterions (Huynh et al., 2016). The mechanism is further supported by the changes of the number of cations bound to the carboxyl groups of PGA in systems under 10 mM cations, which contained one cation in each system (Figs. 7 and S25). For Na^+ and K^+ , there are almost no binding ions around the carboxyl groups of PGA (binding probability = 0.002). Mg^{2+} showed a dynamic process of association and dissociation (Fig. 7A, binding probability = 0.220), which was consistent with the description of Mg^{2+} ions as “diffusely bound” counterions in Manning’s counterion condensation theory (Huynh et al., 2016). Ca^{2+} continuously bound to the carboxyl groups of PGA (Fig. 7B, binding probability = 0.999). Overall, we determined the affinity of cations for carboxyl groups of PGA as: $\text{Ca}^{2+} > \text{Mg}^{2+} > \text{Na}^+ \approx \text{K}^+$ at the same cation concentration.

3.4. Hydration and binding modes of cations to PGA

To understand the effects of solvent on the interaction between cations and PGA, we focused on the interaction between water and cations and calculated the coordination number (CN) of water molecules in the first coordination shells of the cations, which are obtained from the simulations with only one cation in a water box. As shown in Figs. 6D and S26, the CNs of water molecules around Na^+ , K^+ , Mg^{2+} , and Ca^{2+} were about 6, 7, 6 and 7, respectively. Our results were consistent with previous studies (Huynh et al., 2016; Lerbret & Assifaoui, 2022; Piquemal et al., 2006; Wu et al., 2010). For comparison, we calculated the CNs of water molecules around cations of PGA systems in 10 mM salt, for each system containing only one cation. The distance of the first hydration shell of cations was similarly obtained from the RDF of water molecules around the cations (Fig. 6D). We found in these simulation systems that the CNs of water molecules around Na^+ , K^+ , Mg^{2+} , and Ca^{2+} were about 6, 7, 5.6, and 5, respectively. The decrease in the CNs of water molecules around Mg^{2+} and Ca^{2+} in PGA systems compared to in pure water was attributed to the interactions between the cations and PGA, which induced the release of structured water molecules from the hydration shells of the cations (Larson et al., 2023). To elucidate the specific binding mode of cations to PGA, we analyzed the distribution of

cations and water molecules around PGA, and focused on the cations that specifically bound to the carboxyl groups of PGA (Fig. 8). We calculated the distribution of water around these cations in their first coordination shell. As shown in Fig. 8(A–C), when Na^+ , K^+ and Mg^{2+} bound to the carboxyl group of PGA, the CNs of water molecules were 5, 6, and 5, respectively. Compared to the CNs of 6, 7, and 6 water molecules around cations in conditions without PGA, our results indicate that the cations only bound to one oxygen atom from the carboxyl group of PGA (monodentate coordination) (Fig. 6D). However, Ca^{2+} preferentially bound to the carboxyl group of PGA in a bidentate coordination that the CN of water molecules was 5 when Ca^{2+} bound to the carboxyl group, whereas the CN was 7 in pure water (Fig. 6D). Fu et al. (2014) reported similar results that Ca^{2+} typically formed a bidentate coordination with a guluronate monomer. In addition, the six water molecules from the first hydration shell of Mg^{2+} define an octahedron (Fig. 8C), in line with a previous study (Huynh et al., 2016). In contrast, the structure of the hydration shell of Ca^{2+} is much more irregular (Fig. 8D). It is indeed unlikely that the two oxygens from the carboxyl group of PGA, which are separated by an average distance of 2.4 Å, can fit into the octahedral hydration shells of Mg^{2+} (the average distance between the oxygens of two neighboring water molecules is about 2.8 Å), since this would distort rather regular the octahedral first coordination shell of Mg^{2+} (Table S7 and Fig. 8C). Therefore, Mg^{2+} tends to bind to the carboxyl groups of PGA via a monodentate coordination mode. However, the irregular hydration shell of Ca^{2+} (the distances between the oxygens of neighboring water molecules ranging from 2.3 to 2.9 Å) allowed the two oxygens of the carboxyl group in PGA to fit into the hydration shell of Ca^{2+} . These results suggested that, although there was no significant difference in the number of Mg^{2+} and Ca^{2+} around PGA (Fig. 6A), their distinct binding modes might account for the different conformational changes observed in PGA chains.

In order to clarify the mechanism by which Ca^{2+} induces cross-linking via bidentate coordination, we ran simulations of two PGA chains in the presence of 200 mM Ca^{2+} or Mg^{2+} . Two cross-links were formed between PGA chains in the presence of Ca^{2+} (Fig. 9A and B), leading to the formation of PGA dimers and the development of a continuous three-dimensional network, which was consistent with the observed increase in particle size (Fig. 5D). Fig. 9C showed that Ca^{2+} cross-linked PGA chains by binding to carboxyl groups via bidentate coordination. In the case of Mg^{2+} -PGA, cross-linking did not occur because Mg^{2+} could not release water molecules from its first hydration shell (Huynh et al., 2016). Thus, the mechanism of binding between Mg^{2+} and PGA only caused softening of PGA but not cross-linking, which was consistent with previous work showing that the gelation of Mg^{2+} -PGA solutions did not occur owing to weak interactions between the cations and PGA (Donati et al., 2009). These results were also in agreement with the predominantly outer-sphere complexes formed by Mg^{2+} with the phosphate and carboxylate groups of nucleic acids and

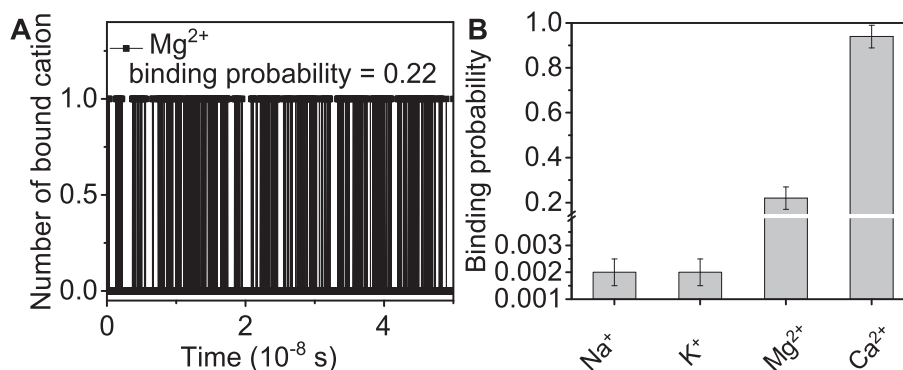


Fig. 7. Binding probabilities of cations on the carboxyl groups of PGA at a concentration of 10 mM. (A) Number of bound Mg^{2+} obtained from MD trajectories. (B) Average binding probability for each cation type, calculated based on the number of bound cations.

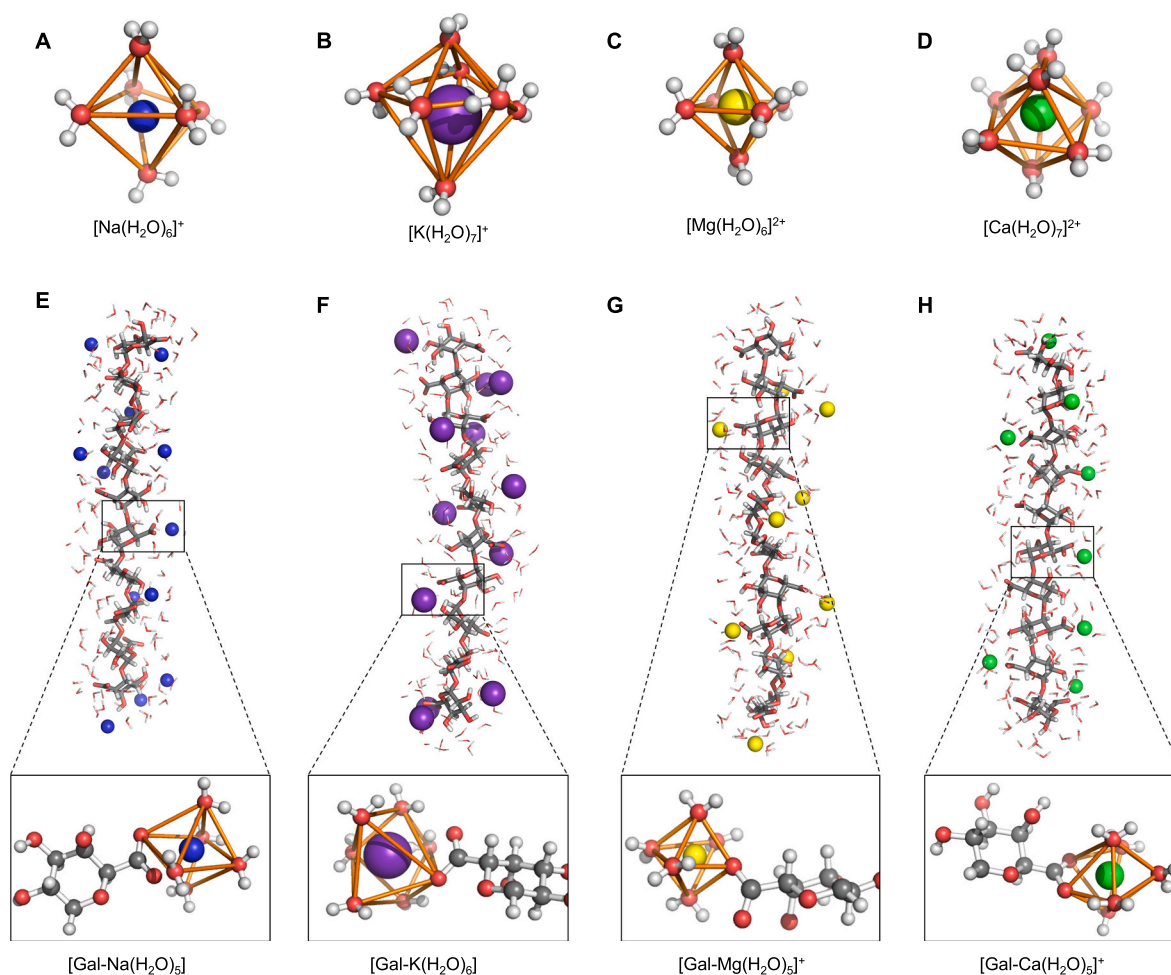


Fig. 8. The snapshots display free cations and bound cations with their water shells. Water molecules were represented by lines in the snapshots and by ball-and-stick models in the inset illustrations. The lines were drawn between the oxygen atoms found in the first coordination shells of cations. The distance of the first hydration shell of cations was obtained from the RDF of water molecules around the cations. (A–D) The free Na^+ , K^+ , Mg^{2+} and Ca^{2+} with their hydration shells. (E–F) The cations binding to PGA with the hydration shells of cations. The calculations were performed at 4 M Na^+ , 4 M K^+ , 0.5 M Mg^{2+} and 0.5 M Ca^{2+} .

proteins, since its affinity for water was higher than that of negatively charged groups (Fu et al., 2014). Our results suggested that the difference in affinity of divalent cations for water molecules drove their different interaction modes with PGA.

In summary, our results revealed that the interaction modes of monovalent cations (Na^+ , K^+) and divalent cations (Mg^{2+} , Ca^{2+}) with PGA. Na^+ , K^+ , and Mg^{2+} bound to PGA in a monodentate coordination mode, while Ca^{2+} bound to PGA in a bidentate coordination mode. It is noted that although Mg^{2+} is a divalent cation, it exhibited weaker affinity for the carboxyl groups of PGA and interacted with PGA as “diffusely bound” counterions (Fig. 7A). Therefore, the mechanism of interaction between Mg^{2+} and PGA was solely based on softening under the experimental conditions examined. In contrast, Ca^{2+} interacted with PGA like a crosslinker. Ca^{2+} was able to coordinate with carboxyl groups and to induce inter-chain crosslinking of PGA due to the high affinity for carboxyl groups, which were verified by the zeta potential and particle size analysis (Figs. 5C–D and 9). However, in the single-molecule MT system, where the PGA molecules were constantly stretched, no cross-linking between PGA molecules occurred when the force exceeded 1 pN, except at 200 mM Ca^{2+} (Fig. 2H). Therefore, the significant decrease of PGA length and P was mainly attributed to the electrostatic screening of cations. The reversal of the P was caused by the charge reversion of PGA under high concentrations (≥ 0.2 M) of divalent cations (Fig. 5A and B). Although Ca^{2+} exhibited a higher affinity for the carboxyl groups of PGA, there was no significant difference in the distribution of divalent

cations (Ca^{2+} and Mg^{2+}) around PGA and also in the binding free energy between divalent cations and PGA when cations-induced cross-linking of PGA were not involved (Fig. 6A and B). Accordingly, the P values showed no significant difference between Ca^{2+} and Mg^{2+} . Similarly, no significant difference in P values was observed between Na^+ and K^+ . These findings offer key insights into the ion-dependent modulation of PGA elasticity and demonstrate how ionic environments influence mechanical properties of PGA. The reversible control of PGA persistence length via ion concentration directly informs the design of pectin-based hydrogels, which are widely used in food texture control, drug delivery systems, and encapsulation technologies. Overall, this work offers mechanistic clarity on how ion-polysaccharide interactions regulate biomechanical behavior in both biological and industrial systems.

4. Conclusions

In this study, we developed a single-molecule assay to directly stretch PGA chains and investigate their mechanical responses under a wide range of cation concentrations. By functionalizing the two ends of PGA, we anchored it between a functionalized glass surface and a superparamagnetic bead, enabling precise measurement of the mechanical properties of PGA. We revealed that the P of PGA was highly dependent on the valence and the concentration of the cation. Specifically, monovalent cations (Na^+ , K^+) and divalent cations (Mg^{2+} , Ca^{2+}) induced a significant decrease in P due to charge neutralization, while

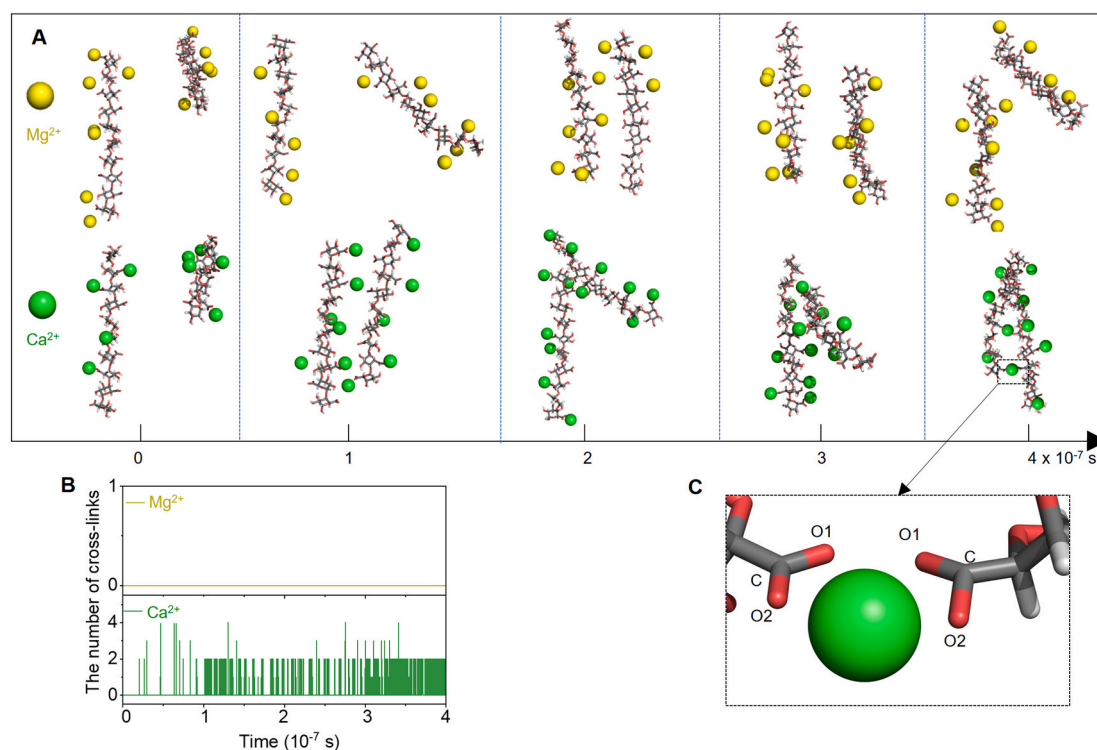


Fig. 9. Simulation results of two PGA chains in the presence of 200 mM Ca²⁺ or Mg²⁺. (A) Snapshots of two PGA chains at different simulation times. (B) Time evolution of the number of cations cross-linking the two PGA chains (C) A Ca²⁺ cation cross-linking carboxyl groups of two PGA chains. The atoms of carboxyl groups is labeled.

high concentrations of divalent cations led to a reversal in P due to net charge inversion of PGA. Molecular dynamics simulations further clarified that Ca²⁺ binds to PGA in a bidentate coordination, while Na⁺, K⁺, and Mg²⁺ interact via monodentate coordination. These results highlight the fundamental role of ionic interactions in regulating the biomechanical properties of PGA. This study provides a novel approach for investigating the mechanical properties of polysaccharides at the single-molecule level and offers new insights for understanding their roles in biological systems, materials science, and bioengineering.

CRediT authorship contribution statement

Yun-Long Chen: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Chen-Chen Zheng:** Writing – review & editing, Writing – original draft, Formal analysis. **Fu-Jia Tian:** Writing – review & editing, Validation, Supervision, Resources. **Tai-Lang Yin:** Supervision, Funding acquisition. **Xing-Hua Zhang:** Validation, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbpol.2025.124268>.

Data availability

Data will be made available on request.

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